

UME Seminar

Some recent high-order solvers with minimal pollution errors for hemodynamic wave propagation

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🔗 [Seminar webpage link](#)

Abstract

Accurate and robust numerical simulation of cardiovascular wave propagation and transport is essential for resolving physiologically-meaningful hemodynamic phenomena across the multiscale human circulatory system. This contribution presents two recently-introduced pseudo-spectral solvers [1, 2] based on a certain numerical framework that invokes high-order trigonometric interpolations of non-periodic functions for the accurate solution of (time-dependent) hyperbolic and parabolic partial differential equations. The first is a space-time solver with minimal numerical dispersion errors for one-dimensional fluid-structure hemodynamics in the entire closed-loop circulation, resolving pressure and flow waves across hundreds of systemic and pulmonary arteries and veins, incorporating nonlinear tube laws, vessel junctions, and fully-coupled lumped-parameter heart, valve, and microvascular bed models. The second is a two- and three-dimensional advection-diffusion solver with minimal numerical diffusion errors for modeling dye concentration evolution (which is linked to drug delivery) and particle residence time (which is linked to thrombus [blood clot] formation), formulated as a post-processing methodology that acts on prescribed velocity fields obtained from either separate numerical simulations or from experimental data. In both cases, the utilization of the same high-order framework enables FFT-speed spatial differentiation, high-order temporal integration, and faithful preservation of the dispersion and diffusion characteristics of the underlying continuous operators, yielding mild CFL constraints and essentially no numerical pollution (errors do not compound over space and time) that is ideal for long distance and long time (e.g., many cardiac cycle) cardiovascular simulations. Performance studies attesting to convergence, stability, and dispersion/diffusion properties are discussed.

This talk also discusses recent applications of these solvers to a range of cardiovascular problems that demonstrate their physiological or clinical relevance and their suitability for patient-specific configurations. For the closed-loop hemodynamics solver, this includes mechanistic investigations of cerebrovascular coupling as well as the robust extraction of clinically-relevant biomarkers (such as those related to heart contractility, ejection fraction, pulse wave velocity, wave reflection, and intrinsic frequencies). For the advection-diffusion solver, this includes simulations of dye evolution and particle residence time (an established marker of thrombus formation) applied to partially-grafted aortic dissections and non-Newtonian Fontan circulations, as well as to catheter-based intraventricular interventions.